

Earlier this year, Ericsson, in partnership with SK Telecom and BMW, disclosed the 5G outdoor mobility trial in South Korea. This included the first multi-vehicular 5G trials. The trials were conducted on a 5G test network that was installed in the car test track of German auto giant BMW in Yeongjongdo city. The South Korean wireless communication major, SK Telecom, along with Ericsson, deployed network slicing and 5G radio network infrastructure to completely cover trial tracks in the BMW driving centre. The trials were completed successfully and showed that 5G can support V2X (vehicular connectivity) services that require low latency and consistently high bidirectional throughput.

The trial also highlighted new key 5G capabilities with multi-site, multi-transmission points with multiple devices operating in the millimeter-wave frequency band.

In terms of machine-type communications, 5G will support massive low-cost IoT connections and with wide coverage, providing connectivity in inaccessible areas as well. This is useful for industries that involve exchange of large amount of data. For instance, in fleet management, a 5G network will ensure smooth connectivity by sending updates on engine performance, total miles driven and faults identified in the vehicle. Thus, it would obviate unnecessary costs, optimise drive time as well as maximise fuel efficiency.

Besides, experts suggest that 5G will soon become the underlying fabric of our ecosystem, connecting all industries with each other. Robotics and automation, artificial intelligence, avionics and agriculture are just a few other industries that will certainly enjoy what 5G has to offer.

Where India stands

Currently, telecom players in India are exploring different upgraded ver-

sions of 4G, such as 4.5G, 4.5G Pro and 4.9G, to improve coverage and add capacity to the existing network. Moving on to 5G, while this technology is set to be launched in 2020, both the government and the private sector have already started work on it.

According to a report, the government has created a research team to do the ground work on 5G in a bid to understand the technology better. It has already filed 100 patents so far. These patents will

A market report suggests that 5G-enabled digitisation revenues in India will be around \$26 billion by 2026. Domestic operators can earn an additional \$13 billion revenues if they go beyond their traditional roles and serve as service enablers

enable India to generate indigenous intellectual property (IP), thus giving bargaining power to the country.

Apart from this, telecom giant Ericsson has signed an MoU with IIT Delhi to jointly dole out '5G for India' programme. The company will set up an incubation centre as well as a 5G test bed to work on the technology. In line with Prime Minister Narendra Modi's Digital India initiative, the programme will help in application development for startups and industries.

Other telecom majors such as Airtel and BSNL have joined hands with Nokia to roll out 5G network in India. The MoU signed between them also covers related infrastructure development.

Similarly, Reliance Jio has partnered with Samsung to bring

5G network to India as well as enhance its existing LTE network. A market report suggests that 5G-enabled digitisation revenues in India will be around \$26 billion by 2026. Domestic operators can earn an additional \$13 billion revenues if they go beyond their traditional roles to serve as service enablers.

However, all this can be achieved only if India successfully overcomes the many challenges in implementing 5G. Significant investment is required to upgrade the existing infrastructure to 5G. Security and privacy too are major concerns.

Increasing the connectivity is yet another challenge. According to Telecom Regulatory Authority of India (TRAI), India had 391.5 million subscribers at the end of December 2016 and a lot of work needs to be done in order to bring everyone into the system. This also implies that consumers will have to upgrade to new smartphones in order to enjoy 5G as this technology is not compatible with previous versions of phones.

Lastly, the biggest concern is the availability of spectrum. 5G requires a new high-frequency band to achieve the desired data speed.

What's next

Experts forecast that once 5G is successfully deployed, it will be upgraded with satellite networks that include telecom satellite, earth imaging satellite and navigation satellite networks.

There are proposals floating around that suggest installation of nano antennae at different geographical locations such as roadsides, villages, malls and hospitals to broadcast high-speed electromagnetic signals. High-speed optical fibres will be introduced to carry high-speed broadband signals. And it seems like sea to space connectivity would be possible once 5G is deployed completely! **EFY**